STOCKWARS
HYU INVESTMENT CLUB

Research Team

Company Analysis
2024.11.05

BH Co Ltd.(090460, KOSPI)

Vanished AppleCar, Found Right here?

Investment Opinion **BUY**, Target Price KRW 33,400

Disappointment over Apple in 2024 has turned into anticipation for the 2025 new-prodect lineup, and doubts over the acquisition of LG's wireless charging business were answered by 2024 earnings. Backed by solid growth drivers, we look forward to BH's golden future ahead!

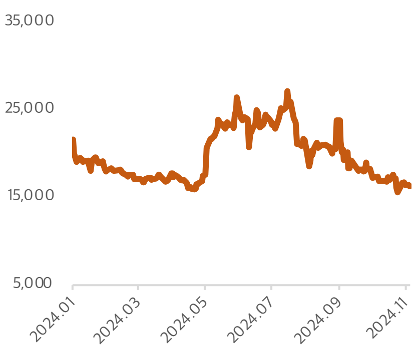
BUY (2024.11.05)

Target P.	KRW 33,400
Current P.	KRW 16,400
Upside	103%

STOCK DATA (2024.11.05)

Market Cap.	KRW 5,625bn
Shares Outstanding	34,464,379
52 Week High	KRW 27,400
52 Week Low	KRW 15,510
Foreign Ownership	13.22%
Major Shareholder	KH Lee 21.14%

Price Trend (2024.01.02~2024.11.05)



1. Apple in 2025 …

Starting with OLED adoption in the iPad, OLED displays are spreading across Apple's product lineup. In 2025, when OLED is applied to every iPhone model, another supercycle can be expected. BH is the main vendor of Samsung Display, Apple's core display supplier, holds overwhelming competitiveness within the Apple pipeline.

2. Another Pipleline: Samsung Electroncis

BH has also performed well in the Samsung Electronics pipeline. While supplying OLED display FPCBs for Samsung smartphones, the company entered this year as a new vendor for smartphone camera modules. Through this, it is building references in camera modules and plans to expand into automotive cameras, XR, robotics, and other fields.

3. New Growth Driver: Automotive Electronics

As affiliates operating automotive-electronics businesses built on BH's FPCBs expand and grow, BH's supply volumes are expected to increase rapidly. These affiliate businesses cover in-vehicle wireless charging modules and battery management systems for EVs and ESS, and are steadily and stably securing large-scale orders from global automakers.

Financial Data (KRW bn)	**2022**	**2023**	**2024E**	**2025E**
Revenue	16,811	15,920	18,360	20,033
Operating Profit	1,313	848	1,392	2,060
OP Margin (%)	7.8%	5.3%	7.5%	10.1%
Net Profit	1,407	849	1,270	1,871
PER	5.3	8.0	7.2	6.8
EPS (KRW)	4,205	2,631	3,687	5,429

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C/O/N/T/E/N/T/S

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1-1. PCB / FPCB Overview

1) What is PCB?

PCB Classification

A PCB (Printed Circuit Board) is a type of electronic board that electrically connects and fixes numerous IT components. Depending on characteristics and applications, PCBs are classified into rigid (RPCB) and flexible (FPCB) types.

RIgid RPCB

An RPCB (rigid PCB) is a stiff, conventional board made from hard insulating materials such as phenol and epoxy resin. Because electronic products can be mounted on it without wobbling, it is used universally. By application, RPCBs are divided into mainboard substrates and IC-substrates that connect the mainboard to semiconductor substrates. However, RPCBs bend poorly and break easily, and multiple points must be connected via wires and cables. These limitations made it difficult to mount them in small electronic devices and the board **developed to solve this problem is precisely the FPCB.**

2) FPCB: The Core of Electronics Compnents

Flexibe FPCB

An FPCB (flexible PCB) is a board in which polyimide (PI), a highly flexible material, is added to the conventional rigid PCB. FPCBs are ① heat-resistant, ② thin and flexible enough to be installed in complex spaces, and ③ able to act as a conductor without multiple connections, shortening work time. Today FPCBs are essential components wherever electronic parts are used smartphones, automotive electronics, medical devices, aerospace and are categorized by structure into **single-sided / double-sided / multilayer / RF types.**

Single/Double/Multi

A single-sided FPCB, with circuit layers on one side only, is the most basic structure; a double-sided FPCB has circuit layers on both sides, raising component density within the same footprint. Multilayer FPCBs connect each circuit layer through PTH (Plated Through Holes) and bond them with adhesive materials to protect the circuits. By stacking multiple FCCL sheets, more circuits and components can be mounted, which is also advantageous for miniaturization.

[Fig. 01] PCB vs FPCB

	PCB	FPCB
Similarities	Replace conventional copper wiring	
	Enables High-density wiring	
	Continuous and automated production	
Differences	3D wiring Unable	3D wiring able
	No bending	Repeated bending O
	Superior Mechenical Strength	Miniaturization and weight reduction
	Easy Multilayer Manufacturing	Greater Design Freedom

Source: Circuitline

[Fig. 02] FPCB

Source: R&D Tech

1-1. PCB / FPCB Overview

Best of both worlds: RF-PCB

The RF-PCB combines the RPCB and FPCB to maximize the strengths of each — a composite board with excellent space efficiency. Rather than joining separate boards, the inner layers are FPCB while the non-bending outer layers are RPCB. In other words, physical strength is maintained where needed while free-form wiring remains possible.

Also, unlike FPCBs, RF-PCBs need no connectors for module-to-module connections, specializing in miniaturization and weight reduction. The conventional signal path of [RPCB–connector–FPCB–connector–RPCB] is shortened to [RPCB–FPCB–RPCB], improving signal transmission speed and reliability. **RF-PCBs suit high-spec component**s, and are mainly used in smartphone displays, automotive electronics, semiconductors, batteries, and camera modules, with application expected to expand into wearables.

Among domestic companies, Young Poong Electronics, BH, Interflex, and Newflex operate in this FPCB business, while Samsung Electro-Mechanics and Daeduck Electronics exited the market as profitability weakened under Chinese price competition, choosing to focus on higher value-added segments.

End market: Set Makers

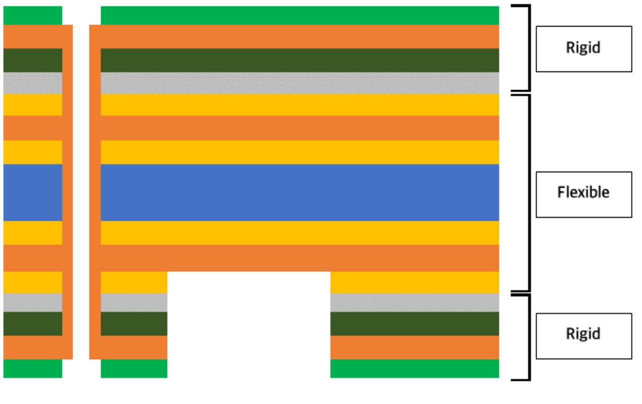
2) Chrateristics of the FPCB Industry

The FPCB industry is a typical build-to-order business whose ultimate target market is set makers producing smartphones, OLED panels, camera modules, and automotive components. When set makers place orders, FPCB firms procure raw materials from upstream suppliers, fabricate parts to customer specifications, and deliver them to the final set maker through module companies.

Set makers weigh three main factors when selecting an FPCB supplier.

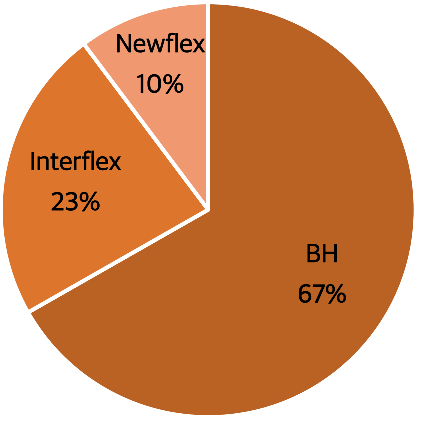
First, the company's references. Because the FPCB value chain is tied to both upstream and downstream industries, past delivery track record and market credibility are key criteria.

[Fig, 03] RF-PCB cross-section



Source: R&D Tech

[Fig. 04] 2024 Major FPCB market player



Source: Connect 007

1-1. PCB / FPCB Overview

Key Factors

① References

② Quality

③ CAPA

Second, quality. Many set makers, including Apple, verify this through rigorous qualification tests. The FPCB manufacturing process runs Roll-to-Roll under physical pressure and temperatures above 270°C, so yields are low at around 50%. Minimizing the probability of thermal deformation of the FPCB film and securing yield is therefore the core of FPCB quality management.

Third, CAPA. The ability to respond flexibly to bulk orders or sudden demand spikes is required, so sustained CAPEX is also a key competitive strength.

FPCB Chicken game of 2010s

Ultimately, set makers select firms with all three factors in balance. This was evident during the rapid FPCB market expansion of the early 2010s: as smartphone penetration drove surging FPCB demand, many companies rushed in with aggressive CAPEX. Leading players — Interflex, BH, Flexcom, and others — over-expanded capacity to digest volumes quickly, and a chicken game began.

Winner's Secret?

But in 2014, as smartphone market growth slowed, FPCB makers faced oversupply and earnings fell under excessive capital spending. Companies that had built factories in Vietnam all at once failed to stabilize yields, took even bigger hits, and went through delisting or bankruptcy. BH, however, managed quality and paced its CAPA — producing core sub-materials in Korea and expanding its Vietnamese plants sequentially — maintaining stable yields and emerging as the winner of the chicken game.

Fast-growing FPCB Market

This FPCB market is expected to grow from USD 21.81bn in 2023 to USD 56.7bn in 2032, a CAGR of +11.2% — roughly twice as fast as the conventional PCB market.

FPCB makers ultimately supply set makers such as Apple and Samsung through the value chain in [Fig. 05], and the key growth engines of this FPCB industry are OLED and automotive electronics.

[Fig. 05] FPCB Industry value chain

Raw Materials & Components

PI Advanced Materials, Innox, Doosan Elec., Hanwha L&C etc.

FPCB

BH, Interflex, Newflex, SI Flex etc.

Display Panel

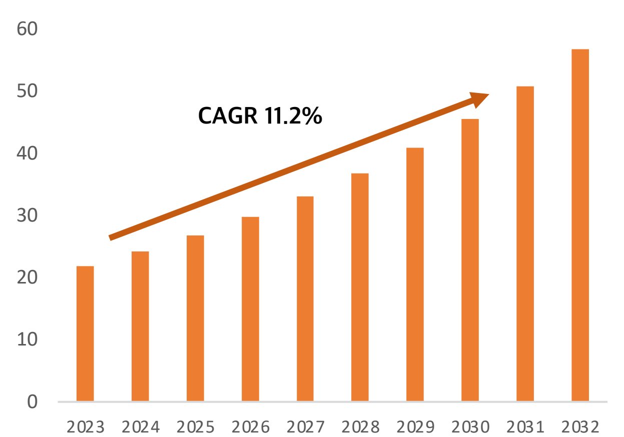
Samsung Display, LG Display, BOE etc.

Finished Sets

Apple, Samsung Electroncis, etc.

출처: Dart, STOCKWARS Research Team 2

[Fig. 06] FPCB Makret Growth Rate (Unit: Billion USD)



출처: Precedence Research

1-2. FPCB's Strong Backers: OLED and Automotive Electronics

1) The Fast-Growing IT OLED Market — with Apple

Expanding OLED display adoption

The display industry expects expanded OLED panel supply on rising IT market demand. According to global research firm Omdia, IT OLED market revenue is forecast to grow from USD 2,534mn this year to USD 8,913mn in 2029, a CAGR of 28.36%. OLED penetration within the IT panel market is expected to reach 37.7% by 2029. As adoption expands alongside technical advances in smartphone OLED, the RF-PCB connecting the OLED panel to the mainboard is structurally bound to grow as well.

Apple is switching to OLED

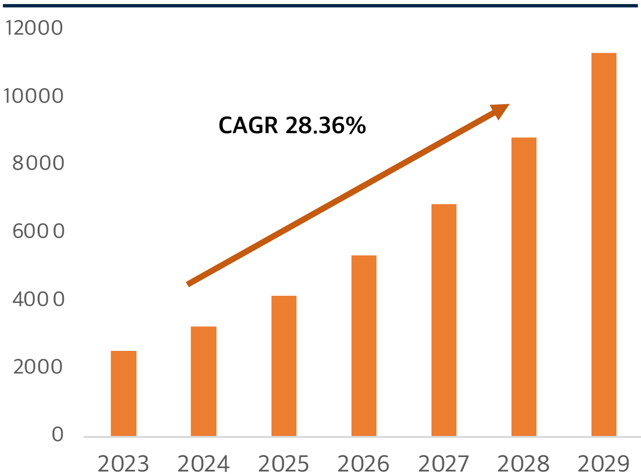
Apple, at the center of this ecosystem shift, first adopted OLED panels in the 2017 iPhone X and has applied OLED to all iPhones except SE models since the 2020 iPhone 12 series. The 4th-generation iPhone SE launching next year will also switch from LCD to OLED, and OLED adoption keeps rising in other devices such as the iPad and MacBook. Conventional LCD and mini-LED displays, relying on backlights, suffered poor dark-color expression and 'blooming.' OLED, by contrast, emits light from each pixel without a backlight, delivering perfect blacks and fitting the slim-and-light trend. Early OLED suffered burn-in, but recent tandem OLED technology has extended lifespan and mitigated the issue.

2) Automotive FPCB Demand Keeps Growing

Automotive FPCB optimized for compactness

With EVs launching in earnest, automakers are focused on lightening internal components to improve efficiency and freeing up space for additional devices through volume optimization. FPCBs have begun to be adopted in automotive cameras, packaging, and battery areas as the device solving these goals, and research applying FPCBs is under way in many other areas. In short, the FPCB's once-limited application scope has expanded in earnest into the automotive-electronics domain.

[Fig. 07] IT OLED Market Revenue Est. (Unit: Million USD)



Source: UB Research, STOCKWARS Research Team 2

[Fig. 08] Apple OLED transition roadmap

	2024	2025	2026	2027
iPhone	iPhone16 Pro (LTPO) iPhone16 (LTPS)	iPhone17 (LTPO)	iPhone18 (UDC) Foldable iPhone	iPhone19 Pro (POL-less)
iPad	iPad Pro (Tandem OLED)		iPad Mini/Air (Single OLED)	iPad Pro (POL-less) Foldable iPad
MacBook		MacBook Pro (Tandem OLED)	MacBook Air (Single OLED)	
Apple Watch	Apple Watch 10 (LTPO)			

Source: UB Research

2-1. Company Overview

1) BH?

Company Overview

Founded in 1999, BH is a dedicated FPCB manufacturer producing and supplying FPCBs core components of advanced IT industries and their applied parts. It also runs an automotive-electronics division responsible for developing and producing in-vehicle wireless chargers. Major customers include large domestic and overseas IT firms such as Samsung Electronics, LG Electronics, Samsung Display, and Apple, with production facilities in Korea, China, and Vietnam. Listed on KOSDAQ in 2007, BH transferred its listing to the KOSPI market in 2023 following a conditional KOSDAQ delisting. The largest shareholder is former CEO Kyung-hwan Lee with a 21.03% stake, while the National Pension Service and treasury shares account for 7.33% and 8.94%, respectively.

2) Governance Structure

Affiliates

The company's governance structure is shown in [Fig. 09]. Its affiliates can be classified into ① BH electronics and BH Flex Vina, in charge of FPCB manufacturing and sales; ② BH EVS Co., Ltd. and BH EVS America, in charge of manufacturing and selling mobile-phone wireless charging modules; ③ DKT and DKT VINA in the display and automotive-electronics segments; and ④ BH Flex HK and Rich Gold Development as other businesses.

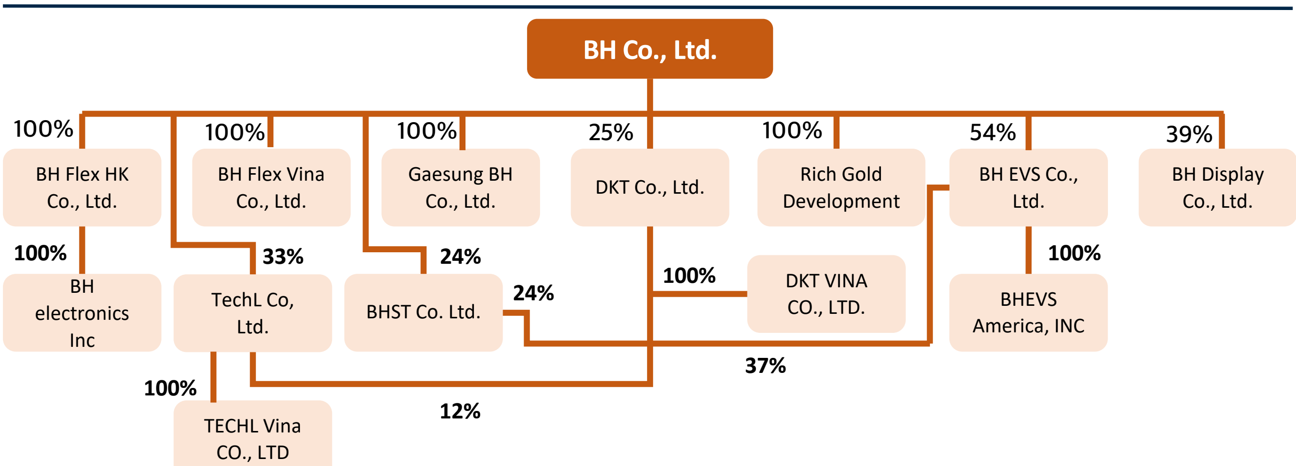
DKT

Briefly on closely related affiliates: DKT is an SMT specialist that receives a portion of BH's FPCB volume and attaches MLCCs, IC chips, and other parts to produce FPCAs. Having acquired LG Electronics' wireless charging business with BH at a 5:5 ratio, it handles contract manufacturing for BH EVS.

TechL

TechL is a back-end SIP specialist focused on securing power-semiconductor technology, particularly automotive power-management chips. Since being acquired by BH in September 2022, it has been working with DKT, BH EVS, and other affiliates to secure growth drivers in new businesses such as automotive power semiconductors and automotive fabless design.

[Fig. 09] Governance Structure



Source: Dart

2-2. Business Divisions

1) FPCB Division

Core is RF-PCB

The FPCB division manufactures FPCBs and their applied components, and is the company's flagship business, generating about KRW 1.2tn, 79.88% of total revenue, as of 2023. BH's FPCBs go into electronics such as mobile phones and PCs, supplying single-sided, double-sided, multilayer, and RF-PCBs. Among these, the RF-PCB accounts for 60% of total company revenue and is the item driving growth. With the slim-and-light trend in electronics expanding the applicable product range, the relevant market keeps growing — and the company grows with it.

Rising Demand, Steady CAPA Expansion

Having won the past chicken game against competitors, BH began growing rapidly by supplying RF-PCBs to Apple alongside the iPhone's OLED adoption, and is set to grow further on the OLED transition across Apple products and other IT devices. The FPCB division is expanding CAPA to match this demand: last October it invested KRW 60bn to expand its Vietnamese production plant, and it committed additional CAPEX in the first half of this year to prepare for steady growth.

2) Automotive Electroncis Divisions

BH EVS

The automotive-electronics division, BH's new growth engine, manufactures and sells through affiliates. It is also expanding into in-vehicle wireless charging solutions and communication fields fusing NFC with wireless charging technology. Through BH EVS, established in October 2022 via joint investment with subsidiary DKT, the company acquired LG Electronics VS division's in-vehicle mobile wireless charging module business for KRW 136.7bn, thereby obtaining Tier-1 supplier status with global OEMs in North America, Europe, and Japan, responsible for design and supply. The division recorded revenue of about KRW 300bn in 2023 (20% of total revenue) and about KRW 186bn in 1H24 alone (24% of total revenue).

2-1) BH EVS & DKT

BH EVS counts existing OEM Tier-1 global automakers, GM, Stellantis, Honda, and others, among its main customers. Until recently, orders were placed entirely on an EMS basis with HLDS, the partner network built by Japan's Hitachi and LG VS, but with affiliate DKT registered as a Tier-2 supplier, DKT now handles part of the volume. DKT was founded in 2012 as an SMT specialist, producing FPCAs by attaching IC chips to FPCBs supplied by BH. In December 2023 it passed product certification from BH EVS vendors and was registered as a Tier-2 supplier, beginning contract production for EVS; in 2024 it established an ES (Energy Solution) unit to prepare to extend its IT technologies into the automotive-electronics domain.

2-3. Financial Analysis

1) Earnings Analysis

Stable Growth Trajectory

Revenue grew sharply from about KRW 650bn in 2019 to about KRW 1.6tn in 2023, and 1H24 posted revenue of KRW 464.8bn (+53% YoY) and operating profit of KRW 31.3bn (+40% YoY). Operating margins were relatively weak in 2020 and 2023 for the following reasons. **[2020]** Samsung Display raised Samsung Electro-Mechanics' share to 50% to secure a stable vendor base, cutting BH's supply volume; this recovered in 2021 when Samsung Electro-Mechanics exited the RF-PCB business as intensifying competition eroded profitability. **[2023]** The impact of weak IT device demand; as the AI market opened in 2024, demand recovered and the operating margin improved to about 7%.

The share price has risen more than tenfold from 2016 to the present. The starting point of this ten-bagger surge was Apple's adoption of OLED in the iPhone 8 in 2016. However, lacking growth drivers beyond Apple's iPhone, the stock has traded sideways over the past few years.

2) Increase in Long-Term Borrowings

Long-Term Deby for CAPA Expansion

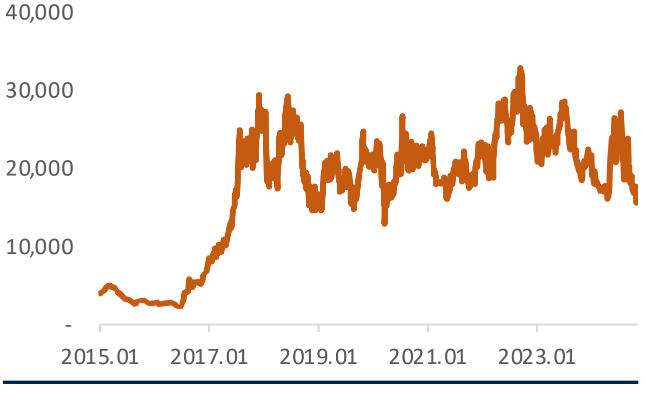
In the first half of this year, liabilities expanded sharply on long-term borrowings (+676%). Confirmation from the IR desk indicates the funds are for the CAPA expansion of Vietnam Plant 4, due for completion in 1H25 — not a result of financial instability or working-capital shortfalls. Despite roughly KRW 55bn of added borrowings, the debt ratio remains sound: 75% in 2022, 67% in 2023, and 95% in 1H24.

3) Cash Conversion Cycle and Inventories

Healthy CCC

The cash conversion cycle (CCC) is the number of days for operating activities to convert into cash; lower values generally indicate healthier finances and stronger competitiveness. BH's CCC was 24.2 in 2021, 19.0 in 2022, and 21.8 in 2023 — low versus peers. Inventories rose modestly last year (+38% YoY), lifting inventory turnover days, but this was driven by work-in-progress and, in our judgment, is far removed from the sluggishness caused by unsold products.

[Fig. 10] Share price 2015~2024



Source: Fnguide, STOCKWARS Research Team 2

[Fig. 11] Key Financial Indicators

	2021	2022	2023
Current Ratio	136%	128%	137%
Debt Ratio	94%	75%	67%
ROA	12%	16%	8%
ROE	23%	29%	14%
CCC	24.2	19.0	21.8
Revenue Growth	44%	62%	-5%
OP Growth	109%	85%	-35%

Source: STOCKWARS Research Team 2

3-1. Apple in 2025 …

iPad OLED display

1) iPad OLED Adoption Complete

The biggest change in the display industry this year is the iPad's adoption of OLED. The 7th-generation iPad Pro, unveiled in the first half, carried an OLED panel for the first time in an iPad, applying three technologies: hybrid OLED, LTPO, and two-stack tandem.

Model coverage set to expand

Sales of the product fell short of expectations, attributed to high prices and a replacement cycle that has not yet arrived. iPad prices ran USD 999 for the 11-inch and USD 1,299 for the 13-inch, and the typical iPad replacement cycle is estimated at five years. Considering iPad demand that recovered from 2020, the need to upgrade to this year's Pro model was limited. Replacement demand from the 2020–2022 cohort is expected to arrive across 2025–2027. In particular, **from 2026–2027 onward, OLED will be adopted sequentially in the value-stream iPad Air and mini models, accelerating replacement demand.**

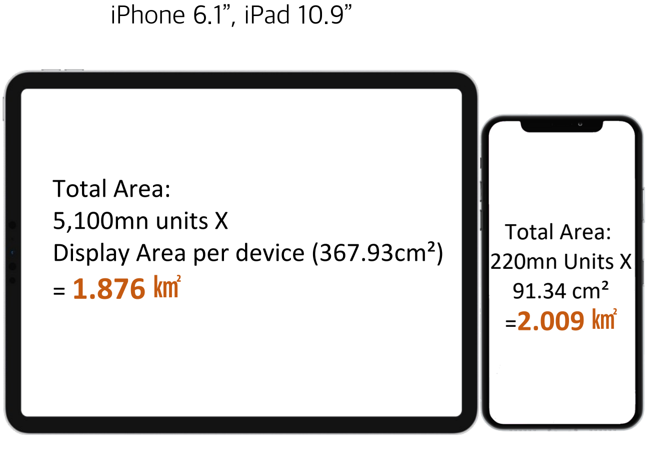
Demand area 4x the iPhone's

Also, while iPad shipments are lower than smartphones', each unit requires a larger panel area. In 2022, annual iPad shipments were 51mn units versus 220mn iPhones, but as in [Fig. 12], the iPad's average panel area is about 4x the smartphone's. In effect, the real demand area is similar — meaning demand area on the scale of the existing iPhone is newly created in iPad OLED panels. Moreover, larger screens and extra circuitry supporting high performance such as M-series chips make ASPs likely to exceed the iPhone's.

iPhone -> iPad -> MacBook

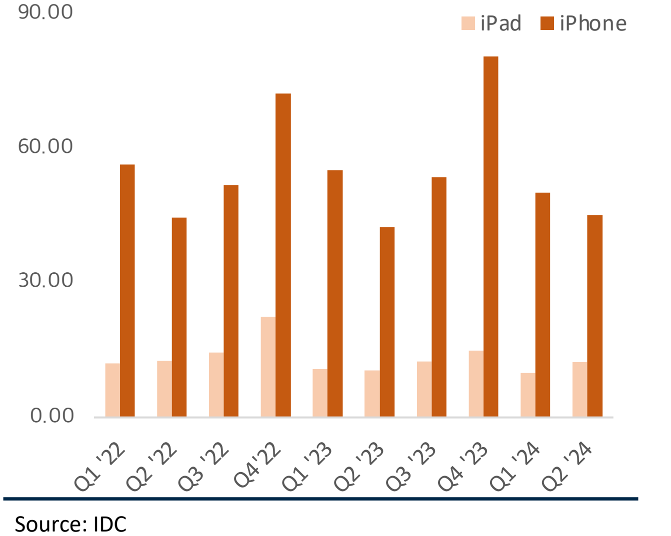
Against this backdrop, Apple's mid-to-long-term roadmap calls for expanding OLED across the product lineup — following the iPhone and iPad, starting with MacBook Pro models in 2026–2027. MacBooks also ship in smaller volumes than iPhones, but given their performance and panel area, we expect P and Q to grow in tandem.

[Fig. 12] '22 iPhone vs iPad shipment area



Source: Hana Securities, IDC, Press Reports

[Fig. 13] Quarterly iPhone/iPad Shipments (Unit: million unit)



Source: IDC

3-1. Apple in 2025 …

2) Expecting a 2025 iPhone Supercycle

What about iPhone 17

Benefits can also be expected from the iPhone, where OLED is applied to every model. iPhone 16 sales this year were weaker than expected, but China sales rose 20% versus the prior model, and with Iran's iPhone embargo lifted and AI features coming, signs of a sales recovery are visible. **<u>The iPhone 17 series launching in 2025 is expected to see explosive demand growth on the full-scale introduction of Apple Intelligence and the arrival of the replacement cycle.</u>**

Apple Intelligence incoming

Currently, the only iPhones capable of running AI are the premium iPhone 15 models (Pro, Pro Max) and the iPhone 16 series; premium iPhone 15 models accounted for just 5.94% of the 1.4bn iPhone installed base as of January this year. During the 2021 iPhone supercycle, Apple restricted the key 5G feature from older iPhones and differentiated with all-model OLED screens and a design change. Apple customers show strong interest in models with new specs, and the introduction of Apple Intelligence is expected to stimulate device replacement demand.

Replacement cycle arriving

The iPhone replacement cycle is also arriving. The iPhone 12 (2020) sold 130mn units and the iPhone 13 (2021) about 100mn — the highest volumes of the past four years. Considering iPhone depreciation and average usage periods, the replacement cycle can be conservatively estimated at 3–4 years. Some replacement demand was expected to flow into the iPhone 16, but we judge that weak sales deferred much of it to the iPhone 17. Notably, CIRP puts Apple customer loyalty at 90%, with churn far lower than rivals — **<u>supporting expectations of a new supercycle in 2025.</u>**

3) BH's Overwhelming Competitive Edge for Apple

To explain BH's competitiveness toward Apple, two value chains flowing **[BH → Samsung Display → Apple]** must be understood.

3-1) Apple Supply Competition: Samsung Display vs BOE vs LG Display

SDC, clear No. 1

Apple, BH's end customer, sources displays from Samsung Display, LG Display, and China's BOE. We explain why SDC maintains its entrenched position despite Apple's vendor diversification policy through a competitor comparison.

① <u>BOE: Late 8.6 investment + unproven deposition equipment + insufficient technology.</u>

Investment later than SDC's

BOE is building an 8.6-generation OLED line with about KRW 11tn announced in December 2023, targeting mass production in late 2026. SDC, by contrast, invested KRW 4.1tn preemptively in April 2023 to convert an existing LCD fab into its A6 plant, and production is expected to start six months earlier. SDC will thus be able to supply 8.6G OLED for part of the iPhone 17, and even if BOE starts production on schedule, SDC holds the advantage through initial iPhone 18 volumes.

3-1. Apple in 2025 …

Sunic System deposition Equipment

Furthermore, BOE adopted deposition equipment from Sunic System, which has almost no track record supplying such tools, whereas SDC uses Canon Tokki equipment already proven through 6G OLED. Canon Tokki can build only two units per year, and with SDC's preemptive orders, supply to others will be difficult for the next year — hampering BOE's efforts to secure performance competitiveness.

LTPO TFT Technology

On technology as well, SDC possesses the LTPO TFT technology applied to premium iPhone and Samsung models, while BOE supplied LTPO OLED samples to Apple last year but approval remains uncertain. Accordingly, SDC is likely to lead supply for the iPhone 17 series, where LTPO TFT is planned for all models. In addition, the Trump administration's second-term China tariff policy contemplates tariffs of 60%+ on all Chinese goods and blocking circumvented imports. If actually implemented, the possibility of BOE's exclusion from Apple's supply chain is considerable; combined with potential US regulation of Chinese displays and patent disputes with SDC, the environment makes it hard for Apple to source large panel volumes from BOE.

② <u>LG Display: still 6G OLED + gloomy finances</u>

Persistent Financial Strain

Amid financial distress, LGD recently secured about KRW 2tn by selling its Guangzhou large-panel LCD plant to China's CSOT, but continues to post large losses. First-half current and debt ratios were 156% and 282%, respectively; despite improving current liabilities through a KRW 1.3tn rights offering in Q1, their scale still reaches about KRW 14.3tn.

LGD therefore plans to defer 8G OLED investment, focusing on 6G alongside debt repayment; even if it converts to 8G, difficulty securing deposition equipment means production would inevitably lag. Some firms still cling to 6G, but 8.6G holds powerful advantages in production efficiency.

[Fig. 14] BOE vs SDC

BOE, Beijing Oriental Technology		SDC, Samsung Display Co.
Announced : Dec 2023 Invest Amount : KRW 11tn CAPA : 32k sheets / month Target MP : late 2026	8.6G Display	Announced : Apr 2023, Sep 2024 Invest Amount : front KRW 4tn, back KRW 2.4tn CAPA : 15k sheets / month Target MP : late 2025
Supplier : Sunic System Reliability : 下, Co- Developed with LGD, unproved	Deposition Equipment	Supplier : Canon Tokki Reliability : 上, 6G Display, proved
MP : 2024, China bound supply Apple bound supply: expected 2026, SDC took 2years	LTPO TFT Display	MP : 2019, Galaxy bound Apple bound supply: from 2021, iPhone 13 Pro

Source: Press, STOCKWARS Research Team 2

[Fig. 15] 6G vs 8.6G Display

	6G OLED	8.6G OLED	Change
Glass Substrate size	1500X1850 mm	2290X2620 mm	Area x 2.25 ↑ Utilization x 1.6↑
Panels per substrate	42	92	Efficiency +119%
Process difficulty & QC	Easy	Difficult	
Cost	Low	High	

Source: Press, STOCKWARS Research Team 2

3-1. Apple in 2025 …

3-2) Samsung Display Supply Competition: BH, SI Flex etc.

BH's overwhelming edge

To benefit from the [Samsung Display → Apple] value chain, a supplier must also be competitive within Samsung Display's vendor pool. SDC's iPhone 16 RF-PCB vendors are SI Flex, BH, and Young Poong Electronics — and BH holds an overwhelming competitive advantage.

Quality and CAPA still stabilizing

SI Flex joined the supply chain hurriedly as the replacement when incumbent second vendor Young Poong Electronics caused defect issues, so its quality and CAPA still need stabilizing. Moreover, it had mass-produced FPCBs specialized for camera modules and only took Apple's qualification test for the first time in 2022, so whether it can sustain stable supply remains unproven. Accordingly, BH delivered most of the initial Q3 volumes this year; with SI Flex's financing constrained by weakened financial stability and a scrapped IPO, large-scale investment looks difficult near term — so BH's share should stay solid.

SDC's main vendor

BH, by contrast, has cemented its position as main supplier even as the vendor structure constantly shifts, winning Samsung Display's Best Partner Award for six straight years. In fact, in 2022 and 2024 the company stably supplied over about 80% of SDC's demand volume.

Steady Growth in Vietnam

Additionally, BH secured stable production lines in Vietnam — advantageous in cost, geography, and policy. It established the BH FLEX VINA entity in Vinh Phuc Province in 2013 and has expanded plants sequentially. Even while the smartphone market slowed and many parts makers cut capex, BH steadily expanded Vietnam-centered capacity and is now adding Plant 4. Vietnam revenue keeps growing, accounting for 73% of total FPCB revenue in Q2.

Well-located Vietnam base

BH's Vinh Yen production base sits very close to main customer SDC's Vietnamese plant. In 2017, when BH began supplying Apple, SDC produced Apple-bound OLED in Vietnam, generating strong synergy. SDC announced plans in the second half to build a KRW 2.4tn OLED display plant in northern Vietnam, signaling continued commitment to its Vietnamese base — a development expected to work favorably for its partnership with BH, a long-standing and geographically close collaborator.

[Fig. 16] Apple – Sansung Display – BH Supply shares

Year	Product	SDC share	BH share	BH Share within iPhone	Share price factor
2020	iPhone 12	78%	50%	39%	X
2021	iPhone 13	70%	55%	38.5%	X
2022	iPhone 14	70~75%	80%	56~60%	BH EVS Founded
2023	iPhone 15	70%	N/A	N/A	BH EVS Loss
2024	iPhone 16	50%	80%	40%	BH EVS Profitable

Source: Press reports, STOCKWARS Research Team 2

[Fig. 17] Vietnam production bases

BHFlex VINA Plant 1	FPCB	
BHFlex VINA Plant 2	FPCB	
BHFlex VINA Plant 3	FPCB for EV battery cables	
BHFlex VINA Plant 4	EV BMS Cable FPCB IT & Automotive OLED FPCB	**(Under Expansion)**
DKT VINA	SMT via DKT	

Source: STOCKWARS Research Team 2

3-2. My Other Pipline: Samsung Electronics

1) Supplying OLED Display FPCBs for Samsung Smartphones

Also Supplying Samsung

BH also handles FPCBs for Samsung Electronics. Samsung shipped 226.6mn smartphones last year, holding 20% of the global market. The Galaxy S24, launched early this year, set a franchise record with 1.21mn pre-orders and posted 13.5mn units sold in Q1. Samsung Display currently supplies 100% of panels for the Galaxy S series including this model, and BH is understood to be supplying a substantial share of SDC's Samsung-bound volume this year.

2) Successful Entry into Samsung Camera-Module FPCBs!

Starting with camera modules, expanding business Areas

This year BH newly entered Samsung's smartphone camera-module FPCB supply ecosystem, supplying FPCBs for internal mounting of camera-module actuators. Camera modules are not a high-margin business, but we read this as a strategy to build camera-module references and diversify a business structure concentrated in displays. Through this, the company plans to advance into automotive cameras, extended reality (XR), robotics, and other areas.

Vendor entry achieved

Camera modules are essential components in smartphones, cars, and smart appliances — technology-intensive products demanding high resolution, miniaturization, low power, and high rigidity. Samsung's smartphone camera-module FPCB supply chain had been an oligopoly of Newflex and SI Flex, but BH newly entered on the strength of references and quality, successfully delivering part of the volume. The company built a dedicated camera-module FPCB line at its Vietnamese plant in Q1 and also supplied Samsung's Galaxy Z Fold 6 released in July. It plans to participate in Samsung's smartphone camera-module FPCB development projects for next year's launches.

Demand rising with multi-camera

Multi-camera adoption in smartphones is growing fast. Consumer preference for high-performance cameras has made dual, triple, quad, and hexa camera phones mainstream, with triple or quad configurations now essential in premium models; multi-camera systems keep advancing to deliver better zoom, low-light shooting, and more. Samsung's flagship Galaxy S23 series carries triple and quad camera systems, and the Galaxy Z Fold 6 that BH supplies also features multi-camera.

Amid this growth, TrendForce forecasts the global smartphone camera-module market to grow at a 9% CAGR from 2024 to 2031. Rising multi-camera adoption will accelerate that growth and positively affect FPCB demand. BH plans to ride this trend, deepening cooperation with Samsung Electronics and expanding its share of the camera-module FPCB market.

3-3. New Growth Driver: Automotive Electronics

As affiliates running automotive businesses on BH's FPCBs expand and grow, BH's supply volumes are expected to rise just as fast. The affiliates' automotive businesses serve in-vehicle wireless charging modules and battery management systems for EVs and ESS.

1) In-Vehicle Wireless Charging (WPC): Now a Core Business, Only Growth Remains

BH held a single pipeline through 2023, but from 2024 it has two.

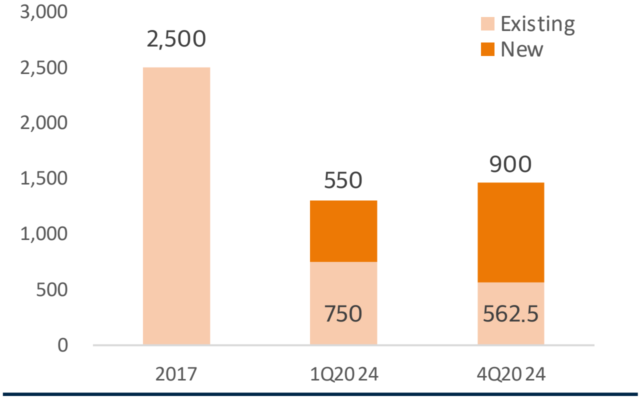
BH EVS

The first pipeline is BH EVS. The purpose of acquiring LG's in-vehicle wireless charging division can be viewed as twofold: ① securing the next-generation in-vehicle wireless charging technology assets held by DKT, and ② securing customer lines. EVS is registered as a Tier-1 supplier to global OEM automakers (GM, Honda, Stellantis, Ford, Renault, Nissan, Volvo, Mitsubishi, Land Rover, etc.), held order contracts of about KRW 2.5tn from 2017, and its remaining backlog is estimated at about KRW 625bn. In 1H24 it signed a KRW 550bn contract to newly supply next-generation in-vehicle mobile wireless chargers (WPC 2.0) to four existing customers from 2026, and in October it added KRW 350bn of orders from European automakers — securing KRW 900bn of orders this year alone. In November it was registered as a Tesla vendor, cementing its No. 1 market share and raising expectations for further orders.

DKT

The second pipeline is DKT, which began quality audits with BH EVS's Tier-1 automaker vendors in early 2023 and passed in December, registering as a Tier-2 supplier. For automotive parts, once a vendor is registered it rarely changes due to safety concerns, so this vendor entry looks like a very positive driver of future revenue growth; roughly KRW 70bn of annual revenue is expected from this year. Also, as BH is registered as the Tier-2 vendor supplying WPC modules for EVs and conventional cars to BH EVS's OEM vendors, Tesla vendor registration is likewise anticipated.

[Fig. 18] Wireless-module order backlog (Unit:KRW bn)



Source: Press reports, STOCKWARS Research Team 2

[Fig. 19] BH EVS Wireless Mobile Charger

Source: BH

3-3. New Growth Driver: Automotive Electronics

Wireless Charging outlook: sunny

According to Strategic Analytics, the adoption rate of in-vehicle mobile phone wireless chargers is expected to rise sharply from 33.5% in 2024 to 44.6% in 2025 and 55.1% in 2026, and accordingly, the potential demand for automotive mobile wireless-charger modules is judged to be very large.

As automotive FPCB applications diversify and customer expansion is anticipated, the company has begun expanding its automotive FPCB production lines. Construction of Vietnam Plant 4 started in 2023 and is expected to enter full-scale operation after completion in the second half of next year. Backed by ample CAPA, aggressive moves to secure new orders and target additional customers are expected.

2) Battery Management System (BMS): The 2024 New Business Enters Full Orbit

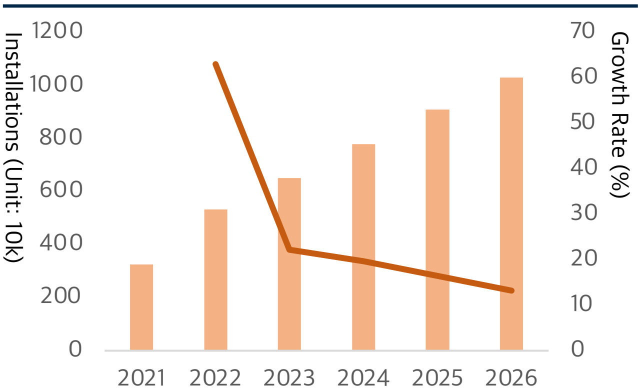
BMS, FPCB is Essential

If 2023's new business was in-vehicle wireless charging, 2024's is the BMS — the control system that monitors battery state and optimizes performance and lifespan. BH runs its BMS (Battery Management System) business through BH EVS, planning mass production of EV battery BMS in 1Q25 and targeting ESS (Energy Storage System) BMS mass production in 1Q26.

BH was the first to develop the in-cell EV battery BMS, with Samsung SDI, SK On, and LGES as current customers. Conventional battery wiring harnesses number 1,500–2,000 per EV, weighing about 50kg. According to the Korea Automotive Technology Institute, replacing them with FPCBs can cut weight by up to 20% and deliver a 0.16% driving-range gain, along with the added benefit of freeing up idle space.

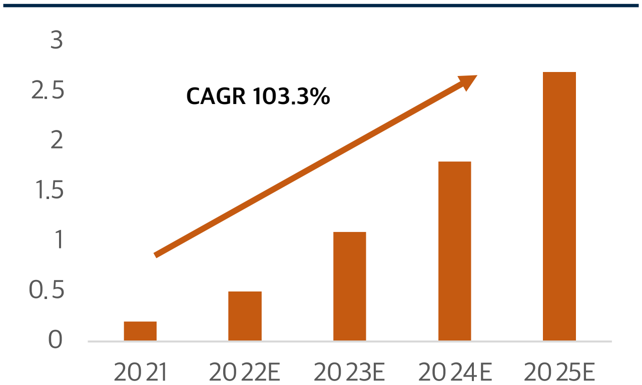
Each EV battery BMS uses 14–15 FPCBs on average, up to 20 in premium vehicles, at an average unit price of about USD 250.

[Fig. 20] wireless charging module installations & growth



Source: Press Reports, STOCKWARS Research Team 2

[Fig. 21] EV battery FPCB Market outlook (unit: Billion USD)



Source:Meritz Securities Research

4-1. Management-Centered Governance, Weak Shareholder Returns

Governance Grade

According to the Korea Institute of Corporate Governance and Sustainability, BH's governance grade was D in 2023 and C in 2024. Grade C is defined as a stage where efforts to build a sustainable-management framework are absolutely necessary and where shareholder value may be impaired by non-financial risks; Grade D is defined as a stage where impairment of shareholder value is a concern. Evaluating BH's 2023 governance through the Korea Productivity Center's K-ESG checklist, board operation appears especially insufficient. Problems include ① the CEO concurrently chairing the board, ② an all-male board, ③ outside directors lacking expertise (no industry-experienced members), and ④ no record of dissenting or amending opinions on board agenda items.

Management-centered board structure

In short, checks and balances on management are severely lacking — and this shows clearly in the executive bonus payment scheme. The company put director compensation limits on the agenda at its 25th AGM following the 24th, and the resolution passed even as the opposition ratio rose from 25.2% to 32.6%. The approved compensation cap is a full KRW 10bn, and since regulations allow bonuses of 0–600% of total compensation, massive incentive payouts remain possible despite shareholder opposition. In the past, single-year bonuses were split across multiple years, making amounts appear smaller, yet KRW 10–20bn per year went to executive compensation — about 10–20% of operating profit. Accordingly, even if earnings grow, the possibility exists that executive pay is prioritized over shareholder returns.

Disappointing Shareholder returns

AGM operation also scores poorly: only electronic voting is in place, and AGM dates are announced just two weeks in advance, tighter than the four-week best practice. Dividend policy is likewise uncertain: citing concerns over earnings declines from slowing end-market growth, the company declined to disclose its future dividend policy. Furthermore, although distributable profit grew substantially after dividends began in 2020, the per-share dividend was fixed at KRW 250, and the payout ratio plunged from 29.81% to 8.73%. This is somewhat understandable given cash outflows from founding BH EVS (KRW 136.7bn) and facility expansion (KRW 60bn), but we judge it difficult to escape criticism of shareholder returns act.

[Fig. 22] BH ESG Grades by Year

	Environment	Social	Governance	ESG Grade
2019	N/A	B	B	N/A
2020	D	C	C	D
2021	B+	B+	B+	B+
2022	C	B+	C	C
2023	B+	A	D	C
2024	B	B+	C	C

Source: Dart, STOCKWARS Research Team 2

[Fig. 23] BH Director Compensation (Unit: KRW 100mn)

Name	Year	Salary	Bonus	Total
KH Lee	2016	9	0	9
	2017	9	0	9
	2018	9.9	34.7	44.6
	2019	9.9	24.7	34.6
	2020	9.6	14.8	24.3
	2021	9.9	14.8	24.7
	2022	9.9	23.1	33
	2023	9.9	17.8	27.7
	1H24	4.9	19.8	24.7
YS Choi	2023	2.5	1.0	3.5

Source: Dart, STOCKWARS Research Team

5-1. Revenue Estimation

1) Apple Bound Revenue Estimation

BH's Apple-bound revenue was estimated by summing iPad and iPhone-bound revenue.

Apple-bound revenue = OLED-equipped product shipments (iPad, iPhone) × panel shipment adjustment (110%) × SDC share × FPCB shipment adjustment (110%) × BH share within SDC × ASP × FX rate

1-1) iPad-Bound Revenue Estimation

OLED panels are slated for iPad Pro models only through 2025, so we set the average share of Pro models in total iPad shipments over the past five years, about 15%, as the OLED adoption ratio through 2025. From 2026, as OLED extends to the iPad Air and mini models at launch, FPCB demand is expected to rise, and we judged it likely that some volume will be pre-ordered in 2H25. Accordingly, we estimated 50% — the share excluding standard models (50% of total shipments) — as the OLED adoption ratio and partially reflected it in 2025 revenue.

Currently, SDC and LGD each supply 50% of iPad OLED panels, and considering LGD's utilization rate and financial condition, we judged that SDC will maintain its 50% share in 2025 as well. The company has supplied 80% of SDC's Apple-bound volume over the past five years, but we conservatively used 66% — the three-year average — as its share. iPad FPCBs are estimated at 2.5-4x the iPhone FPCB unit price (about $9), given that iPad screens are roughly 4x larger and support high-spec chips such as the M4. The estimated iPad-bound revenue is as follows

(Unit: KRW bn)	3Q24E	4Q24E	1Q25E	2Q25E	2024E	2025E
iPad OLED shipments(10k units)	0	139	155	180	471	887
Panel shipments (10k units)	0	153	171	198	518	976
SDC share within Apple	50%	50%	50%	50%	50%	50%
SDS shipments (10k units)	0	76	85	99	259	488
FPCB shipments (10k units)	0	84	94	109	285	537
BH share within SDC	80%	80%	80%	66%	80%	70%
BH shipments (10k units)	0	67	75	72	228	368
FPCB ASP($)	23	23	23	23	27	27
Exchange rate (KRW/dollar)	1,350	1,350	1,300	1,300	1,350	1,300
Estimated iPad-bound Rev.	**0**	**21**	**22**	**21**	**71**	**110**
Total Apple-bound Rev.	296	365	268	190	1,165	1,227

1-2) iPhone Bound Revenue Estimation

OLED is currently applied to all iPhones except the entry SE model; since the SE accounts for about 10% or less of total shipments, the 2024 OLED adoption rate was set at 90%. But with OLED panels scheduled for all models from end-1Q25, we raised the 2025 adoption rate to 100%.

5-1. Revenue Estimation

Through this year's iPhone 16, LTPO technology was applied only to the top two models, but from the iPhone 17 series it is planned for all models. As a result, supply from BOE — whose LTPO capability remains unverified at Apple — is expected to shrink, while SDC's and LGD's shares expand. SDC's supply share in this year's iPhone 16 was about 50%, down from the 70%+ average it maintained over 2020–2023. As compensation for this decline, SDC is expected to supply a small volume to next year's SE model as well (SE model — BOE is the main supplier).

Although SDC's supply is expected to expand in the iPhone 17, we conservatively determined the overall iPhone-bound supply share based on the 2024 iPhone 16 share plus 10% of SE model volume. Accordingly, the supply share is projected to decline in 2Q25 when the iPhone SE is released, but recover to over 60% after the iPhone 17 launch in 3Q25. The estimated iPhone-bound revenue is as follows.

(Unit: KRW bn)	3Q24E	4Q24E	1Q25E	2Q25E	2024E	2025E
iPhone OLED shipments(10k units)	5,040	5,850	4,338	4,097	18,603	20,777
Panel shipments (10k units)	5,544	6,435	4,772	4,507	20,463	22,855
SDC share within Apple	50%	50%	50%	44%	50%	70%
SDS shipments (10k units)	2,772	3,218	2,386	1,983	10,232	12,651
FPCB shipments (10k units)	3,049	3,539	2,624	2,181	11,255	13,916
BH share within SDC	80%	80%	80%	66%	80%	66%
BH shipments (10k units)	2,439	2,831	2,100	1,440	9,004	9,552
FPCB ASP($)	9	9	9	9	9	9
Exchange rate (KRW/dollar)	1,350	1,350	1,300	1,300	1,350	1,300
Estimated iPhone-bound Rev.	**296**	**344**	**246**	**168**	**1,094**	**1,118**
Total Apple-bound Rev.	296	365	268	190	1,165	1,227

2) Samsung-Bound Revenue Estimation

Samsung-bound revenue consists mainly of Samsung smartphone OLED FPCB revenue and camera-module FPCB revenue, and was estimated using the same methodology as Apple-bound revenue. Until last year, OLED panels were applied to Galaxy flagship models (S, Z series), while the high-volume mid-to-low-end A series carried OLED only in some models (A30 series and above).

This year, however, OLED extends to the A1 series and the adoption scope is expected to gradually widen. Accordingly, we used 58%, last year's adoption rate plus the shipment weighting of the A1 series, as the OLED adoption ratio, and held this rate constant through 2025. Currently, SDC is estimated to hold the largest supply share within Samsung's OLED smartphones, and the flagship models are likely to maintain an SDC-centered supply structure. However, as OLED penetrates price-sensitive entry-level lines such as the A series where price competitiveness matters, there is also the possibility that Chinese display makers may expand their share

5-1. Revenue Estimation

Accordingly, we adjusted SDC's supply share downward. In addition, with Interflex and Newflex added as SDC vendors for Samsung-bound supply, we also slightly trimmed the company's share to reflect the possibility of existing vendor share dispersion. For the FPCB unit price, we used industry estimates that reflect the fact that low-end models account for a roughly 50–60% larger share of shipments than Apple's.Camera-module FPCB revenue was substituted with industry estimates, as reference data is still lacking.

The estimated results are as follows.

(Unit: KRW bn)	3Q24E	4Q24E	1Q25E	2Q25E	2024E	2025E
Galaxy OLED Total Rev.	33	48	54	54	191	218
Camera Module Total Rev.	12	7	10	10	19	39
Total Samsung-Bound Rev.	**45**	**55**	**64**	**64**	**210**	**256**

3) Automotive Segment Revenue Estimation

The automotive segment reflects BH EVS's wireless charging module business and DKT's EV division revenue.

Global auto sales rose about 12% in 2023 per Statista, and EVS recorded steep revenue growth. However, 3Q24 vehicle registrations fell QoQ in North America (-0.3%), Europe (-21%), and Germany (-17%), showing that 2024 auto sales momentum is slowing. Our team therefore applied the 2.5% average of research institutions' auto sales growth forecasts and a 5% wireless-charger module market penetration rate to 2023 total revenue to assume 2024 total revenue; for 2025 we applied the same growth rate plus a limited reflection of newly recognized orders

(Unit: KRW bn)	3Q24E	4Q24E	1Q25E	2Q25E	2024E	2025E
BH EVS	82	83	96	105	349	385
EV BMS	9	15	17	17	49	70
Total Automotive Rev.	**90**	**98**	**114**	**122**	**398**	**455**

Some revenue will gradually be recognized as additional vendors register, but as exact order sizes and timelines cannot currently be confirmed, it was not reflected.

(Unit: KRW bn)	2020	2021	2022	2023	2024E	2025E
Total Revenue	**721**	**1037**	**1681**	**1592**	**1,836**	**2,004**
Apple bound Display	453	645	1217	985	1,165	1,228
Samsung bound Display	170	229	242	167	191	218
Camera Module	0	0	0	0	19	39
Automotive	0	27	102	372	398	455
Antenna Cable	26	58	53	31	21	25
Smartphone Battery	0	57	52	31	25	2
Others	0	22	15	6	17	13

5-2. Cost Estimation

1) COGS and SG&A Estimation

The main items within the company's COGS and SG&A are raw-material costs, employee wages, depreciation, and outsourced processing costs. The core item within raw-material costs is CCL, which accounted for 37% of total raw-material costs in the first half of this year; CCL's raw materials are further classified into copper and PI. Copper has been fluctuating this year on expectations for the global economy and China's economic recovery. After copper prices surged in the first half, they have recently turned downward again, but given the uncertainty of the presidential election, we judged it inappropriate to reflect the current trend on a long-term basis. Accordingly, 3Q24 raw-material costs were calculated using the quarterly average share of revenue in 2022, a year that showed a downward trend, while 4Q24 and 2025 raw-material costs were estimated by reflecting the average share of revenue in years when copper volatility was high. PI, the other raw material, and its raw material PMDA have recently shown a downward trend, but unlike copper, for which futures trading is active, the price movement trend cannot be accurately identified, and so was not considered.

Employee wages were calculated by reflecting the labor-cost increase from 4Q bonuses, the headcount increase from 2025 plant expansion, and the minimum-wage growth rate. Depreciation was calculated by applying the trailing one-year average and was adjusted upward from 2Q25 to reflect the 147% CAPA expansion. The company's outsourced processing cost is the SMT outsourcing cost; as it was folded into other costs through 3Q22, it was calculated using the quarterly average share of revenue from 4Q22 onward. Other items of lower materiality were calculated using one- to two-year averages.

2) Non-Operating Items and Icome Tax Estimation

Interest costs, being fairly consistent quarterly within each year, were estimated at the one-year average; FX gains were adjusted with FX movements considered conservatively (KRW 1,400 for 2024, KRW 1,350 for 2025). For income tax, the historical average effective rate of 13% was applied.

(Unit: KRW bn)	2020	2021	2022	2023	2024E	2025E
Revenue	721	1,036	1,681	1,591	1,836	2,004
COGS	658	936	1,481	1,436	1,451	1,669
Gross Profit	63	100	192	155	383	334
SG&A	28	29	61	70	79	128
Operating Profit	34	71	131	85	139	206
Non-Operating Income	(170)	33	33	12	12	9
Pre-tax Profit	33	104	164	97	152	215
Income Tax Expense	7	22	23	12	25	27
Net Profit	25	82	140	85	127	187

5-3. Target PER Valuation

We applied the PER method for the company's valuation. We judged it to be the method that can best explain both the earnings of the existing FPCB segment, driven by the OLED transition in IT devices, and the growth potential of the new automotive segment simultaneously.

An appropriate multiple is difficult to find in the company's historical data. Some similarity (new market creation via expanding applications) can be found at the point of the iPhone's past OLED adoption, but it differs somewhat from the current revenue structure. Since acquiring BH EVS in 4Q22, the company has seen a sharp rise in revenue not only from display-bound FPCBs but also from automotive-bound FPCBs, and is preparing its leap to a core business segment.For similar reasons, we judged that peer companies are absent as well. The company holds overwhelming references and shares in Apple and Samsung-bound FPCB supply, and in the automotive segment as well, it has achieved the No. 1 market share in the automotive wireless-charger module supply market, backed by steady orders from global customers.

Accordingly, Research Team 2 derived a Target PER by applying different weights to each business segment. In 2025, the revenue mix by business segment is expected to be 74% for the IT segment (display, camera module) and 24% for the automotive segment. Considering that IT demand is gradually recovering, we applied an average 12-month forward PER of 7.1 the average during IT upcycles to the FPCB segment, and an average PER of 6.0 for major global finished-vehicle OEMs to the automotive segment. Weight-averaging the PER of each business segment by the company's revenue mix, we derived a Target Multiple of 6.8.In sum, the target price was derived by multiplying 2025 estimated EPS of KRW 5,429 by the Target Multiple of 6.8. However, given the current period of high political uncertainty, we applied a 10% discount to reflect the possibility of a slowdown in the automotive industry's growth. The finally derived target price is as follows.

PER Valuation	
Net Profit (Unit: KRW mn)	187,100
Shares Outstanding	34,464,379
2025E EPS (Unit: KRW)	5,429
Target F/W PER	6.8
Discount Rate	10%
Target Price	KRW 33,400
Current Price	KRW 16,440
Upside	103 %

Target Price and Opinion

Research Team 2 derived a target price of **KRW 33,400,** representing **103% upside** versus the current price of KRW 16,440.

Investment Opinion BUY

Expect the target price to be realized

2H25 ~ 1Q26